Research map of IISc Bangalore and TIFR Mumbai, 2016–2026

DPCN Research Mapping Project · Group 18 · Integrated report · OpenAlex data collected 13–15 Oct 2026

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Integrated dashboard: `dashboard/index.html` (13 tabs; IISc / TIFR / both; flagged papers included or excluded). Public baseline: https://sohamjahagirdar.github.io/DPCN_A2/; this integration is not published.
Code, data files and methods: https://github.com/SohamJahagirdar/DPCN_A2 · deliverables in `outputs/`: group18_data.csv, citation_network.csv, authorship_network.csv, indian_connections.csv

1. Data and scope

All journal articles, reviews, letters, book chapters and conference papers from 2016–2026 with an author at IISc Bangalore or TIFR Mumbai, from OpenAlex: 42,725 papers (845 joint), 49,487 dataset rows. TIFR means Mumbai only (NCBS, NCRA, ICTS, TIFR Hyderabad, TIFR-CAM and HBCSE authors are excluded by affiliation text). Faculty come from the official rosters matched to OpenAlex authors; author lists cut at 100 by the API were fetched in full. Journal quality uses SCImago quartiles in the publication year; impact uses OpenAlex 2-year mean citedness. Full methods: dashboard *Method* tab and the repository README.

	IISc	TIFR	Both
Papers 2016–2026	35,818	7,752	42,725
Citations	610,269	223,983	803,550
Institutional h-index (2016–2026 papers)	217	162	247
Journal papers in Q1 journals (SCImago)	69.0%	71.7%	69.6%
Faculty with papers	720	159	879
Papers with another Indian institute	42.0%	56.3%	43.1%
Papers with > 12 authors (instructor's flag)	1,698 (4.7%)	2,112 (27.2%)	3,119 (7.3%)
Citations lost without the flagged papers	15.2%	71.5%	27.9%
Papers in Nature Index journals	2,345	1,375	3,207
Papers used outside academia in India	27.3%	23.6%	25.9%
Papers cited by patents (published up to 2023)	2.6%	0.8%	2.4%

2. Key statistics

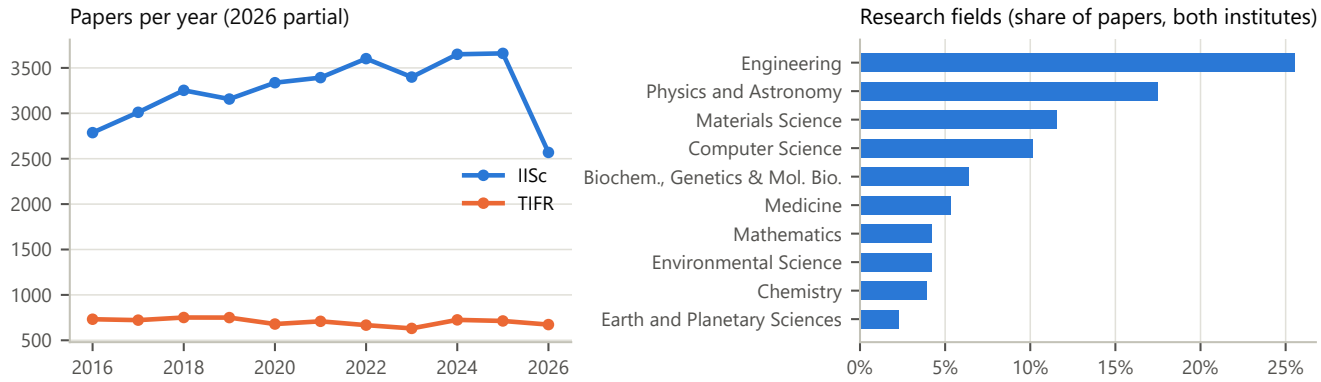


Figure 1. Output per year and research fields. IISc publishes about five times as many papers as TIFR and is engineering-led; TIFR is concentrated in physics and astronomy.

- **Two research models.** 27.2% of TIFR papers have more than 12 authors (large physics and astronomy collaborations), against 4.7% at IISc. Without these flagged papers TIFR loses 71.5% of its citations and IISc 15.2%; the dashboard repeats every analysis without them, as the instructor asked.
- **Journal choice.** 69.6% of journal papers are in Q1 journals, but only 5.8% in high-impact journals (impact ≥ 10). In a logistic regression on faculty–paper pairs, team size is the strongest driver: papers with 13–50 authors have 4.8× the odds of a high-impact journal, other factors held fixed. Assistant professors have 1.34× the odds of professors; papers with other Indian institutes 0.74×; TIFR 0.20× (its main physics journals have impact below 10, a field effect).
- **Reputable journals.** 3,207 papers (10.0% of journal papers) are in the 178 Nature Index journals; they are 37.9% in the global top 10% of their field, against 14.9% for our other papers.

3. Network analysis

3.1 Citation network (papers citing papers)

42,725 papers and 75,581 citations among them; 46.5% of papers sit in the largest weakly connected component, and 53.5% are never cited by another of our papers. The in-degree distribution is heavy-tailed: a discrete power law fitted by maximum likelihood (Clauset et al. 2009) gives $\alpha = 3.06 \pm 0.14$ for papers with ≥ 21 internal citations (209 papers) — a few papers become the internal reference points. PageRank (Brin & Page 1998) ranks them on the dashboard. With the rest of Indian science the flow is balanced: other Indian institutes cite our papers 123,348 times and we cite theirs 128,309 times (ratio 0.96).

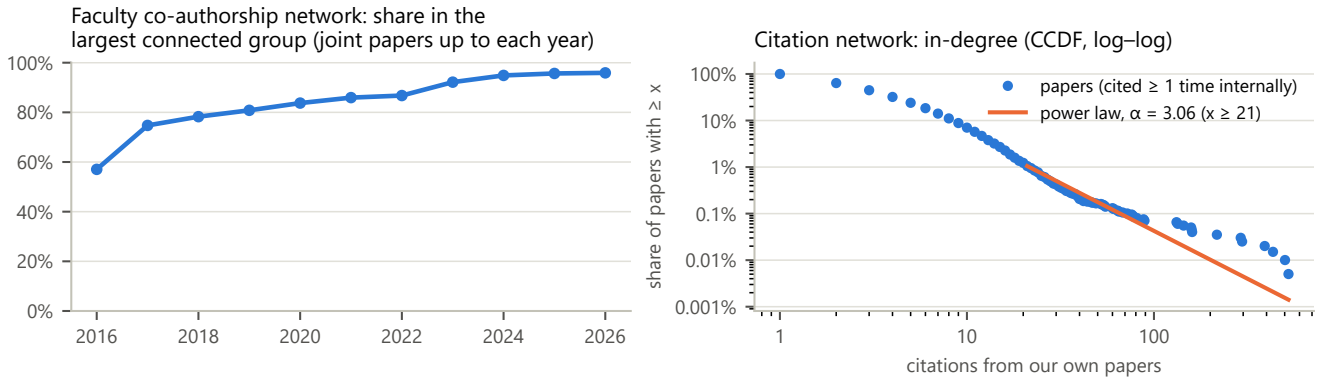


Figure 2. Left: growth of the faculty co-authorship network. Right: complementary cumulative distribution of internal citations with the fitted power-law tail.

3.2 Faculty co-authorship network: more than chance?

879 faculty, 1,814 links (joint papers with ≤ 50 authors), the kind of collaboration network first studied by Newman (2001). To go beyond description we compare the network with 20 degree-preserving random rewirings (Maslov & Sneppen 2002), in which every faculty member keeps their number of co-authors but not their partners.

Measure (708 faculty with a faculty co-author)	Observed	Random	Reading
Clustering coefficient	0.257	0.022	12× random
Mean shortest path (largest group)	5.06	3.84	small world, σ = 8.9
Louvain modularity (unweighted)	0.588	0.427 ± 0.004	z = 41
Links within a department	40.2%	3.8%	10× expected
Degree assortativity	+0.21	-0.03	hubs link to hubs
Rich club (faculty with > 17 co-authors)	0.211	0.142	1.48× random

- **A small world organised by departments.** Clustering is 12× random while distances stay short (σ = 8.9; Watts & Strogatz 1998; Humphries & Gurney 2008). Louvain finds 37 communities (weighted modularity 0.74; Blondel et al. 2008), far stronger than in random networks, and 40.2% of links stay inside a department (expected 3.8%).
- **The two institutes barely connect.** Only 69 of 1,814 faculty links join IISc and TIFR (institute assortativity 0.80), although 845 papers are joint: 691 of these have more than 12 authors, so the institutes meet mainly inside large collaborations.
- **Hubs work with hubs.** Positive degree assortativity (Newman 2003) and a rich club (Colizza et al. 2006): the best-connected faculty collaborate with each other more than chance, and a few bridging faculty (highest betweenness, on the dashboard) hold departments together. The largest connected group grew from 57% of linked faculty in 2016 to 96% by 2026.

3.3 Connectivity with Indian institutions

IISc co-authors with 1,372 Indian institutions and TIFR with 445; the largest partnership is TIFR–Indian Institute of Technology Madras (1,246 shared papers). 43.1% of papers have another Indian institute (TIFR 56.3%, IISc 42.0%). Partners are grouped by the course clusters, with topic similarity (Jaccard over research subfields) for the main partners. A journal co-usage network (journals linked by shared faculty, Louvain) shows the publication ecosystems.

4. Network resilience and global citation reach

4.1 Who keeps the collaboration network connected?

Each faculty researcher is removed independently from the recorded co-authorship graph. We count pairs of **other, surviving researchers** that were connected before removal but no longer have a path afterwards. Pairs involving the removed person are excluded from both the loss and its denominator. Joint papers with more than 50 authors are already absent from both graphs.

Graph	Faculty	Links	Components	Largest	Maximum pair loss
All papers	879	1,814	185	679	6,713 (2.92%)
Without >12-author papers	877	1,642	198	660	21,840 (10.07%)

The highest proportional removal impacts are **Sitharam T G** with all papers and **Raghavan Varadarajan** without flagged papers. Their losses are divided by 229,522 and 216,835 previously connected surviving pairs, respectively. The filtered graph has fewer alternative paths and its worst measured loss is larger.

> **Many collaborators and structural importance differ.** Ambarish Ghosh has the highest all-papers degree (40 collaborators), but a different researcher has the largest removal effect. A low-degree bridge can separate groups that a high-degree hub does not. This is a hypothetical topology experiment, not a prediction of departures, lost productivity or researcher quality.

Institution selection filters researcher rows, while graph-wide measures remain for the combined IISc-TIFR graph. Excluding >12-author papers changes the graph itself. The native dashboard provides rankings, role definitions, CSV exports and selected-researcher fragments.

4.2 Follow the Idea: where are the papers cited next?

The original 42,725 IISc/TIFR seed IDs resolve to 42,719 queried IDs; 6 remain provider-unresolved. The separate frozen OpenAlex study (cutoff 2026-10-15; processed 19 Oct 2026) records **555,786 distinct direct citing works (G1)** and **807,678 seed-citer links**. One work can cite several seeds, so links and works are different totals. Combined unique-citer counts come from citation-pair sets, never sums of seed-level unique counts.

Recorded direct links, both institutions / all seeds	Count
Different primary OpenAlex fields / both fields known	184,420 / 806,331 (22.87%)
Excluded: unknown citing field	1,339
Additional excluded: known citing field, unavailable seed field	8

Thus 807,678 - 1,339 - 8 = 806,331 classifiable links. Cross-field links measure a change of **primary field between papers**, not unique cross-field people, research quality or proven idea adoption.

The institution control selects seeds (joint papers belong to each institution and count once in Both); excluding flagged papers also selects **seeds only**, without deleting downstream citers by author count. G2 comprises selected, bounded second-hop paths and is not a full-corpus second-generation census. Evidence paths and edge provenance remain inspectable in the dashboard.

> **Coverage and evidence limits.** Exhausted provider queries cover available OpenAlex Core records, not all world scholarship. Unknown fields remain excluded from the cross-field denominator. Recent seeds have less observation time: 15,779 resolved seeds lack the study's full five-year window; 26,940 are eligible. Source pages/excerpts may be unavailable; citation evidence does not establish adoption, practical benefit or causality. Independent human source reviews: **0**. Full definitions and case-specific limitations: `docs/FOLLOW_THE_IDEA.md` and `docs/NETWORK_RESILIENCE.md`.

5. Impact on India: importance and relevance

Reputable papers reach Indian research most, but India is a small part of their audience. Measured with Indian citations (self-citation excluded):

	Nature Index	...India-focused	Other papers
Papers	3,207	103	28,909
In global top 10% of field	37.9%	29.1%	14.9%
Cited at least once in India	65.0%	82.5%	55.6%
Indian citations per paper	6.0	19.2	3.4
Indian share of all citations	9.7%	24.8%	17.8%
Cited by a patent (to 2023)	3.1%	0.0%	2.2%
Used outside academia in India	37.5%	74.8%	28.9%

Beyond research. We traced documented pathways from papers to users outside academia (Bornmann 2013): 25.9% of papers are co-authored or cited by Indian companies, government agencies, hospitals or NGOs; 732 papers are cited by 1,411 patents (Reliance on Science; Marx & Fuegi 2020; patent citations are the standard link between science and technology, Ahmadpoor & Jones 2017); 1,348 papers have co-authors at 350 companies in India, including 23 start-ups based at IISc. Atomic energy, health, CSIR laboratories and the weather and climate agencies are the heaviest users.

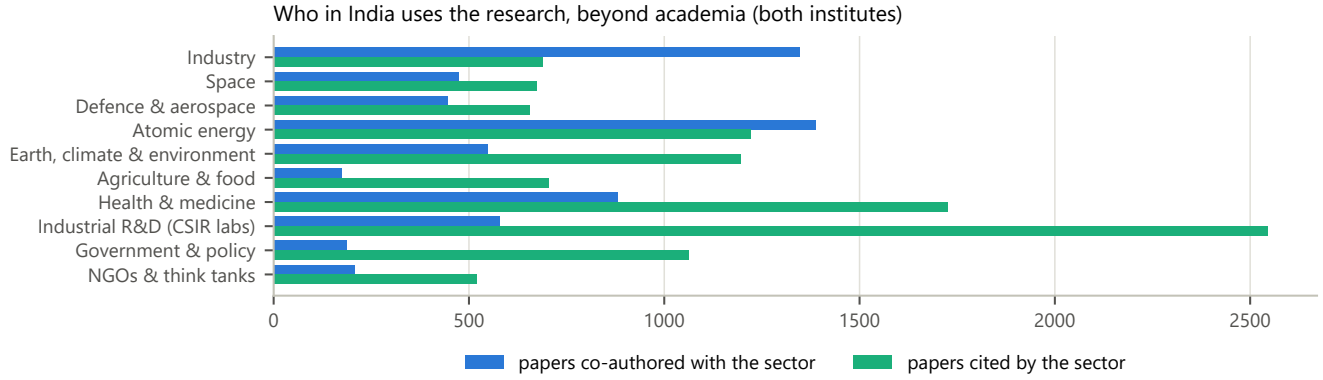


Figure 3. Papers co-authored with, and papers cited by, Indian organisations in each sector (citing papers involving our own institute excluded).

Documented outcomes. 23 case studies with public sources and matching papers where available; the 15 with a documented outcome are below (all 23, with sources, on the dashboard's *Impact on India* tab). Some outcomes have no matching paper in our corpus; case inclusion is not causal attribution:

- **Oxygen concentrators for COVID-19, licensed to 24 companies** (IISc)
- **Hereditary cancer gene tests from an IISc-founded company** (IISc, 12 papers)
- **An ICMR-approved COVID-19 test kit from an IISc start-up** (IISc)
- **A heat-stable COVID-19 vaccine from an IISc start-up** (IISc, 17 papers)
- **Gallium-nitride chips made in India** (IISc, 9 papers)
- **Electromagnetic-compliance software from an IISc spin-off** (IISc, 59 papers)
- **A reconfigurable processor from IISc research** (IISc, 22 papers)
- **MEMS chip manufacturing from an IISc spin-off** (IISc, 4 papers)
- **Industry-funded research: the Robert Bosch Centre for Cyber-Physical Systems** (IISc, 301 papers)
- **Vaani: an open speech dataset of India's languages** (IISc, 21 papers)
- **IISc hosts the National Quantum Mission's quantum-computing hub** (IISc, 63 papers)
- **TIFR-designed qubits in DRDO's 6-qubit quantum processor** (TIFR, 20 papers)
- **Instruments for India's first astronomy satellite** (TIFR, 150 papers)
- **IISc wetland studies used by the National Green Tribunal to protect Bengaluru's lakes** (IISc, 13 papers)
- **The national framework states use to plan climate adaptation** (IISc, 8 papers)

6. Key observations and insights

- **Separate large collaborations.** TIFR's 27.2% flagged papers carry 71.5% of its citations. Author count does not establish individual contribution or invalidate the work.
- **Separate venue indicators from quality.** Q1 is common; high-impact-journal publication is associated with team size. These are observational associations, not causal team effects.
- **Collaboration is departmental.** Short network paths coexist with strong departmental structure and relatively few direct IISc-TIFR faculty links.
- **Two routes to national reach.** Reputable work reaches Indian research; India-focused papers also reach mission agencies and state universities. Documented licences, start-ups and mission roles illustrate practical pathways.

7. Modelling approaches: used and proposed

Used. PageRank on the citation network; Louvain communities; degree and betweenness centrality; degree-preserving null models for small-world, modularity, assortativity and rich-club tests; maximum-likelihood power-law fit; logistic regression with paper-clustered errors for journal choice; with/without-flagged sensitivity analysis. Also implemented: independent single-node removal tests; full-seed direct-citation reach, primary-field transitions and bounded G2 evidence paths.

Interpretation limits. 2026 is partial; journal impact is not field-normalised. The power-law fit is a descriptive tail fit, without a bootstrap goodness-of-fit or alternative-distribution test; it does not establish a unique generating mechanism. Twenty null rewirings are a finite comparison. Patent coverage ends in 2023, excludes the Indian Patent Office and lacks assignees. Registry/address matching undercounts companies. Citation/co-authorship and selected documented cases are pathways, not causal estimates of national benefit, revenue, jobs or policy impact.

Proposed for the second round (state of the art in the science of science; Fortunato et al. 2018).

Disruption indices would compare team sizes (Funk & Owen-Smith 2017; Wu et al. 2019); **block models/Leiden** would test communities (Peixoto 2014; Traag et al. 2019); **temporal/multilayer networks** would connect collaboration and citation histories (Holme & Saramäki 2012; Kivelä et al. 2014). **Link prediction/node2vec** would explore future partnerships (Liben-Nowell & Kleinberg 2007; Grover & Leskovec 2016), and **citation dynamics** future reach (Wang et al. 2013). These remain proposals.

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